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A framework enabling LLMs into regulatory environment for transparency and trustworthiness and its application to drug labeling document

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ABSTRACT

Regulatory agencies consistently deal with extensive document reviews, ranging from product submissions to both internal and external communications. Large Language Models (LLMs) like ChatGPT can be invaluable tools for these tasks, however present several challenges, particularly the proprietary information, combining customized function with specific review needs, and transparency and explainability of the model's output. Hence, a localized and customized solution is imperative.

To tackle these challenges, we formulated a framework named askFDALabel on FDA drug labeling documents that is a crucial resource in the FDA drug review process. AskFDALabel operates within a secure IT environment and comprises two key modules: a semantic search and a Q&A/text-generation module. The Module S built on word embeddings to enable comprehensive semantic queries within labeling documents. The Module T utilizes a tuned LLM to generate responses based on references from Module S. As the result, our framework enabled small LLMs to perform comparably to ChatGPT with as a computationally inexpensive solution for regulatory application.

To conclude, through AskFDALabel, we have showcased a pathway that harnesses LLMs to support agency operations within a secure environment, offering tailored functions for the needs of regulatory research.

1. Background

Document review is the routine task for all the regulatory agencies to evaluate the safety, efficacy, and quality of consumer products. For example, reviewing the documents of Investigational New Drug applications (INDs) and New Drug Applications (NDAs) is one of the critical responsibilities in the U.S. Food and Drug Administration (FDA) to clear a drug for use in the U.S. Moreover, during the review process, regulatory agencies also generate a lot of documents such as guidance documents, internal communications, meeting minutes, etc. This review process heavily relies on manual reading, which is time-consuming, labor-intensive and, above all, expert dependent. This creates two challenges: (1) how to improve the review efficiency and (2) how to maintain the institutional memory for consistency in policy. For the latter, the challenge is not just accuracy but also the issue of knowledge transfer. Manual reading often depends on expert knowledge, making it challenging for others to replicate the work. In addition, the past documents are often referenced and referred during the review process to ensure consistency and evidence-based decision-making in review of a new product.

Recently, with large language models (LLMs) such as ChatGPT, Bard and Llama have shown impressive capabilities in understanding and generating human-like text (Haleem et al., 2022; Koga et al., 2023; Yunxiang et al., 2023). For example, ChatGPT, or the back-end model named GPT-3.5/GPT-4, has shown close to or even better performance than humans in many Natural Language Processing (NLP) domains (Guo et al., 2023). Therefore, many agencies including FDA have explored the possibility of integrating LLMs into the regulatory process but must overcome met at least three challenges.

1. Review-centric functionalities: Many review processes adhere to predefined procedures and queries depending on the nature of review

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requirements. This process often involves proprietary information that are not in the public domain and thus unlikely well represented in the public LLMs. For example, reviewers often need functionalities to support internal ID search which is not readily available from the foundation models. Consequently, a customized function with specific review requirements must be developed alongside the foundational models.

2. Transparency and explainability: This is critical in interpreting the model's output to gain certainty and confidence. Specifically, reviewers need a more transparent chain-of-thought and traceable evidence from the reliable data sources. Concerns may arise among FDA reviewers and researchers regarding the trustworthiness of the model, especially when the model is not transparent (Liesenfeld et al., 2023). One solution is to link the model's output to the original text.
3. Data security: Most regulatory data are sensitive and cannot be put through public LLMs. Specifically, some text data used in regulatory application are not publicly accessible due to confidentiality or privacy considerations. Hence, a localized solution is imperative.

To address these challenges, we developed a Retrieval-Augmented Generation (RAG)-based framework (Lewis et al., 2020) to enhance the customization and transparency of LLMs. The proposed framework was also implemented in a local, secured IT environment, to meet the requested security needs of many regulatory documents.

RAG combined the strengths of large language models with dynamic information retrieval, allowing for the integration of the most current and relevant data into the decision-making process. By developing the RAG module, the input prompt of LLMs could be enriched and customized to meet the unique requirements of regulatory application. Another important advantage of this framework is its ability to integrate future LLMs to stay current with evolving technology in the field. We demonstrated this framework on FDA drug labeling documents that are an essential part of the FDA drug review process.

One of the critical regulatory activities is the examination of current drug labeling documents for knowledge acquisition, precedence investigation, and referencing to support regulatory decision-making. Drug labeling serves as a comprehensive source of information for healthcare professionals and patients, providing details about a drug's indications, dosages, contraindications, and potential adverse effects. Reviewers need to extract relevant information and answer important questions. Currently, there are two main computational solutions to levitate these regulatory tasks involving manual reading; one is to use keywords-based search tools such as DailyMed (National Library of Medicine, 2023) and FDALabel (Fang et al., 2016, 2020) while the other is focused on semantic search such as RxBERT (Wu et al., 2023) and PharmBERT (ValizadehAslani et al., 2023). In this study, we present a localized LLMs-based tool called AskFDALabel. AskFDALabel aims to address the specific needs and challenges of processing drug labeling documents in the reviewing process. By leveraging the power of rapid advancement in LLMs, AskFDALabel offers a more efficient, accurate, and reliable solution for extracting relevant information from drug labeling documents. When extended to the diverse set of regulatory documents, this localized LLMs tool will significantly advance regulatory science with automated assistance, enhancing productivity, and improving the overall review process.

2. Methods and materials

2.1. Overall framework

The framework of AskFDALabel is developed based on RAG with a specific data retrieval function to search and query over current labeling documents. Its overall framework is depicted in Fig. 1, which consists of four major components: document processing, query processing, LLM inference, and user interface. Two model engines were developed in the AskFDALabel, the semantic search of query processing unit (Module S) and the text-generation of the LLM inference with a given prompt (Module T). Given the rapid advancement in AI and LLM technologies, a cassette-swap method was implemented where the models in these two modules can be replaced on demand.

2.2. Labeling data collection and processing

In total, 2705 labeling documents were retrieved from FDALabel (Fang et al., 2016), as only Human Prescription drugs of NDAs and

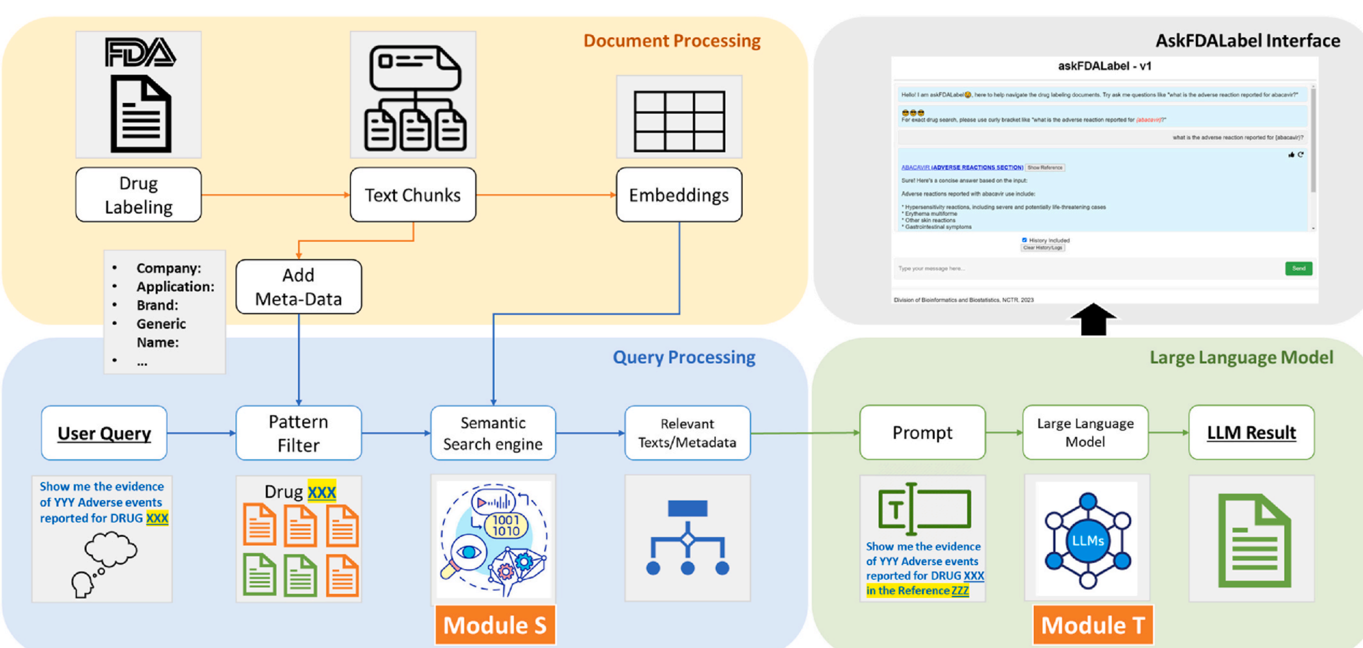

Fig. 1. AskFDALabel Framework. AskFDALabel contains four components: (a) Document processing, (b) Query processing, (c) LLM inference, and (d) User interface. The main purpose of Document and Query processing is to establish semantic search a query which subsequently forms the prompt for LLM to enhance the transparency and trustworthiness of the generative output.

Reference Listed Drugs (RLDs) were included in this study. We further removed Repacker and Relabeler labeling if applicable. The labeling documents were then split into text chunks up to 2000 characters in length using the *RecursiveCharacterTextSplitter* in Langchain (langchain, 2023). As a result, 56,974 text chunks were generated from the 2705 labeling documents. Note that the splitting methods and the chunk size may affect the performance of Module S and further investigation may be needed.

2.3. Module S

The vector store used in the semantic search module was based on sentence embeddings. The Sentence Transformer package (Thakur et al., 2020) was used in this study to convert texts into numerical vectors. Among all of the pretrained models, we selected "all-mpnet-base-v2" since it reported the state-of-the-art performance compared to other models (SBERT. Sentence, 2022). After processing all text chunks, each chunk was converted to a numerical vector of 768 dimensions. During the semantic search, we used cosine similarity to measure the similarity between the query and database vectors. The top candidates with the highest similarity would be considered as relevant texts, which would be used as reference in Module T.

We developed an "optimized" search method (i.e., Search Type = "Optimized"), for the semantic search function in Module S. This method was compared with the conventional sematic search (i.e., Search Type = "Regular"). Specifically, we added metadata onto the end of each text chunk so that the keywords from the metadata would also contribute to the vector. The current metadata applied in AskFDALabel includes the representative organization, product name, generic name, and the application number. The metadata would be added to the end of every text chunk as additional information like the following: "*Additional Information: 'Represented Organization [repOrg]. Product Name [prodName]. Generic Name [genrName]. Application Number [applNum]*".

In addition to the incorporation of meta-information, a stringent keyword-based filter would be employed to narrow down the search results. Specifically, when a keyword pattern aligns with the query, it triggers constraints on the search outcome. For instance, for the query "What are the adverse events of Abacavir?", recognizing "Abacavir" as a distinct product name instructs the semantic search to focus solely on text segments labeled with Abacavir as the product. Users will be informed whenever such filtering is applied in generating the results.

2.4. Module T

2.4.1. Pre-trained models

In the current study, two foundation LLMs were investigated.

- Gpt4-x-alpaca-13b: it is a fine-tuning LLM model (*chavinlo/gpt4-x-alpaca*) based on the foundation model Llama-13b, and further fine-tuned with GPT-4 generated training dataset (HuggingfaceHub, 2023). This pre-trained model was retrieved from HuggingFace hub and deployed in our in-house IT environment with 8 A100 GPUs.
- Falcon-40b (Almazrouei et al., 2023): it is a recently published LLM by Technology Innovation Institute (TII) which was trained on the RefinedWeb Dataset. It contained one trillion tokens from filtered and deduplicated web-curated corpora (Penedo et al., 2023). At the time of access, Falcon-40b-instruct was listed near the top of the open LLM leaderboard (HuggingFace, 2023) in regard to overall performance (Average of 63.4) over the collection of benchmarks. Similarly, the Falcon-40b-instruct model (*tiiuae/falcon-40b-instruct*) was retrieved from HuggingFace hub (HuggingfaceHub: faclon, 2023) and deployed in our in-house IT environment.

2.4.2. LoRA fine-tuning: Dataset

We fine-tuned the pretrained text-generation model, i.e., the Gpt4-x-alpaca-13b and Falcon-40b-instruct models using a similar approach introduced by Alpaca (Taori et al., 2023). In brief, alpaca-style of fine-tuning used an existing strong LLM (such as ChatGPT) to automatically generate instruction data, and used the generated instruction-response data to fine-tune a relatively weak and smaller LLM. In this study, an own-generated training dataset was generated by GPT-3.5 to train both gpt4-x-alpaca-13b and Falcon-40b models, which was formatted in the below style.

```
–
### Instruction:
[Instruction]
### Input:
[Input]
### Response:
[Response]
–
```

Here, [Instruction] is the designed question template with a specified entity name, [Input] is the search result from the semantic search module, and [Response] is the generated text from ChatGPT. In detail, in order to generate the training dataset, five templates were designed based on the different types of common questions that the reviewer may query. These are as follows.

1. **'Warnings and precautions'**: 'find the most serious warnings and precautions of … ',
2. **'Adverse reactions'**: 'what are the adverse reactions of …',
3. **'Dose'**: 'tell me the dose of …',
4. '**Indications**': 'what is the indication of …',
5. **'Drug interactions**': 'what drugs interact with …'

In all, we curated 18,670 samples for the Low-Rank Adaptation of Large Language Models (LoRA) fine-tuning process. It is worth noting that these five templates may not cover all possible questions that may be answered by the reviewers. In other words, the model may be further tuned with a more comprehensive training dataset.

2.4.3. LoRA fine-tuning: Training

Regarding the fine-tuning process, the Parameter-Efficient Fine-Tuning (PEFT) package from HuggingFace was used for the LoRA fine-tuning. Particularly, we fine-tuned four types of weights ("query_key_value", "dense", "dense_h_to_4h", "dense_4h_to_h") from the original Falcon-40b architecture. For the Gpt4-x-alpaca-13b pre-trained model, we fine-tuned two weights ("q_proj", "v_proj"). The other parameters used in the LoRA fine-tuning process are listed below: (*lora_r = 8, lora_alpha = 16, lora_dropout = 0.05, batch_size = 128, micro_batch_size = 4, gradient_accumulation_steps = 32, learning_rate = 1e-4, train_steps = 500, task_type = "causal_lm"*).

2.5. Computational resources

The data processing and fine-tuning process of AskFDALabel was performed on an in-house GPU server, with 8 Nvidia A100 GPU cards and by using CUDA-11-7. The total GPU memory was 640G (80 Gb * 8). For the LoRA fine-tuning process, all models were loaded and fine-tuned in a quantization way (8-bit).

The fine-tuned adapter for the Gpt4-x-alpaca-13b model (*chavinlo/gpt4-x-alpaca*) has been uploaded to HuggingFace Hub and is available at https://huggingface.co/seldas/AskFDALabel_v1_lora.

3. Results

3.1. Module S performance

One important measure to assess search relevance was based on whether the query result originated from the correct labeling documents. We developed three experiments to mimic the real review usage scenario, involving search on three different types of entities – generic name, product name, and application number. For example, the query could be in the form of "*Tell me the adverse reactions of Lipofen?*" or "*Tell me the adverse reactions of NDA022006?*". The output of module S was then compared to the ground truth – which was obtained by FDA-Label database query of the designated entities. The result was considered a match if the output chunks of module S was originally from the true label. The overall relevance was measured by the ratio of matched cases to total tested cases. For each type of entity, around 2000 cases were tested.

The performance of both the "regular" and "optimized" strategies in Module S is summarized in Table 1. Clearly, using the "optimized" strategy, which involved applying a keyword-based filter for a specific entity during the semantic search, yielded significantly better results compared to the "regular" semantic search. This method proved especially effective when searching for application number, as the "regular" semantic search often struggled to locate most of the relevant results, possibly due to the limitations of tokenization when dealing with numbers. Consequently, the utilization of a keyword-based filter effectively addressed this issue.

3.2. Module T LoRA fine-tuning training

Domain-specific fine-tuned model may help to improve the inference performance, as current foundation models are pre-trained with general domain texts and their adaptation to labeling related questions may be limited. For that, we applied LoRA, which freezes the pre-trained model weights and injects trainable rank decomposition matrices into each layer of the Transformer architecture, to fine-tuning the models for downstream tasks. Specifically, the LoRA approach was applied to fine-tune both the Gpt4-x-alpaca-13b and Falcon-40b pretrained models using the PEFT package. Out of the 18,670 ChatGPT (GPT-3.5) generated data samples, 16,670 were allocated for training, while the remaining 2000 were used for validation during the PEFT training process. Each pretrained model (Gpt4-x-alpaca-13b and Falcon-40b) was loaded in 8 bits to reduce the resource requirements and fine-tuned for 500 steps. The fine-tuned model was evaluated at every 50 steps using the validation dataset. The training and validation loss for the Llama and Falcon training is displayed in Fig. 2. In general, we observed a flattened training curve after 400 steps, indicating that the training process reached a saturation point at this stage. Additionally, we noted that the training and validation losses exhibited similar and consistent trends.

3.3. Module T LoRA fine-tuning evaluation

After LoRA training, we compared the LoRA model to the base model for Gpt4-x-alpaca-13b and Falcon-40b, respectively. GPT-4 was used as the judge to compare the performance between the LoRA fine-tuned

Table 1
Comparative analysis with three typical usage scenarios in regulatory application.

Entity	Search Type	Relevance	Improvement (over Regular)
Generic name	Regular	24.9% (462/1859)	
Generic name	Optimized	85.1% (1582/1859)	+60.2%
Product name	Regular	48.8% (1073/2201)	
Product name	Optimized	95.0% (2091/2201)	+46.2%
Application Number	Regular	0.05% (1/2153)	
Application Number	Optimized	99.16% (2135/2153)	+99.11%

models and their corresponding base models. For this comparison, there were 4 categories of results: (1) **LoRA** is good (2) **Base** (i.e., without LoRA training), is good, (3) **Both** are good, or (4) **None** (i.e., neither answer is good enough to answer the question based on the reference). In most cases, **None** meant the Q/A model failed to capture important or relevant information from the reference, or the reference did not contain relevant information.

We used 100 random product names and application numbers to conduct this evaluation, on both the Gpt4-x-alpaca-13b and Falcon-40b based models. As shown in Fig. 3, we observed: (1) for the Gpt4-x-alpaca-13b model, LoRA could significantly improve the model performance (i.e., "LoRA is good" was much larger than "Base is good"); (2) for the Falcon-40b model, LoRA did not improve the performance; (3) both Gpt4-x-alpaca-13b and Falcon-40b showed a similar proportion of "None" category; and (4) the performance for product names and application numbers were similar.

It is worthwhile to point out that we used ChatGPT generated answers for the fine-tuning of both LIama-13b and Falcon-40b. Since Falcon-40b may already have similar power/capability as ChatGPT, the fine-tuning may not be very helpful. However, for Gpt4-x-alpaca-13b based model, fine-tuning with ChatGPT generated answers in the "Alpaca" style can significantly improved the smaller, localized model performance (Taori et al., 2023). This offers an opportunity to use a less expensive solution to carry out some tasks in an environment where the computational resource is limited.

3.4. Comparison between LIama-13b and Falcon-40b in module T

Besides comparing the LoRA and based model of the same foundation model, we also compared the performance between Gpt4-x-alpaca-13b and Falcon-40b. We randomly selected 100 product names and application numbers to evaluate the model inference process. For each product name (or entity), we posed a question such as "what are the adverse reactions reported for [entity]," and the answer was generated based on the top three selected text chunks.

We compared the inference results of Gpt4-x-alpaca-13b and Falcon-40b with GPT-4 serving as the judge. To elaborate, for a given question such as "What are the adverse reactions of NDA021543?", the semantic search provided the reference context, and the Q/A module employed either the Gpt4-x-alpaca-13b or Falcon-40b model to generate responses based on the question and reference independently. Subsequently, we presented both responses, along with the question and reference materials, to GPT-4 for assessment to determine which answer was better. For example, GPT-4 might provide feedback such as "Answer 1 is good. The second part of Answer 1 and Answer 2 are not related to the question". In general, these responses can be categorized into four groups: (1) **Llama** is good, (2) **Falcon** is good, (3) **Both** are good, or (4) **None** (indicating that neither answer is satisfactory for addressing the question based on the reference). In most cases, "**None"** implied that the Q/A model failed to capture important or relevant information from the reference, or the reference did not contain relevant information.

We conducted this evaluation using 100 randomly selected product names and application numbers for both the Gpt4-x-alpaca-13b and Falcon-40b fine-tuned models (Fig. 4a). For comparison, we also used the no-LoRA versions of the of Gpt4-x-alpaca-13b and Falcon-40b models (Fig. 4b). As illustrated, following LoRA fine-tuning, the category labeled "None" exhibited a significant reduction for both the Gpt4-x-alpaca-13b and Falcon-40b models, suggesting that the fine-tuning process may contribute to the improvement of the final response. Additionally, we observed a notable preference for the Gpt4-x-alpaca-13b based model when compared to the Falcon-40b based model, particularly in the comparison of the LoRA versions. Lastly, the patterns between product names and application numbers exhibited similarities.

Based on the previous results, we noticed that LoRA fine-tuned Gpt4-x-alpaca-13b can improve the performance, whereas for Falcon-40b, no-LoRA seems better. Therefore, we then compared these two "best"

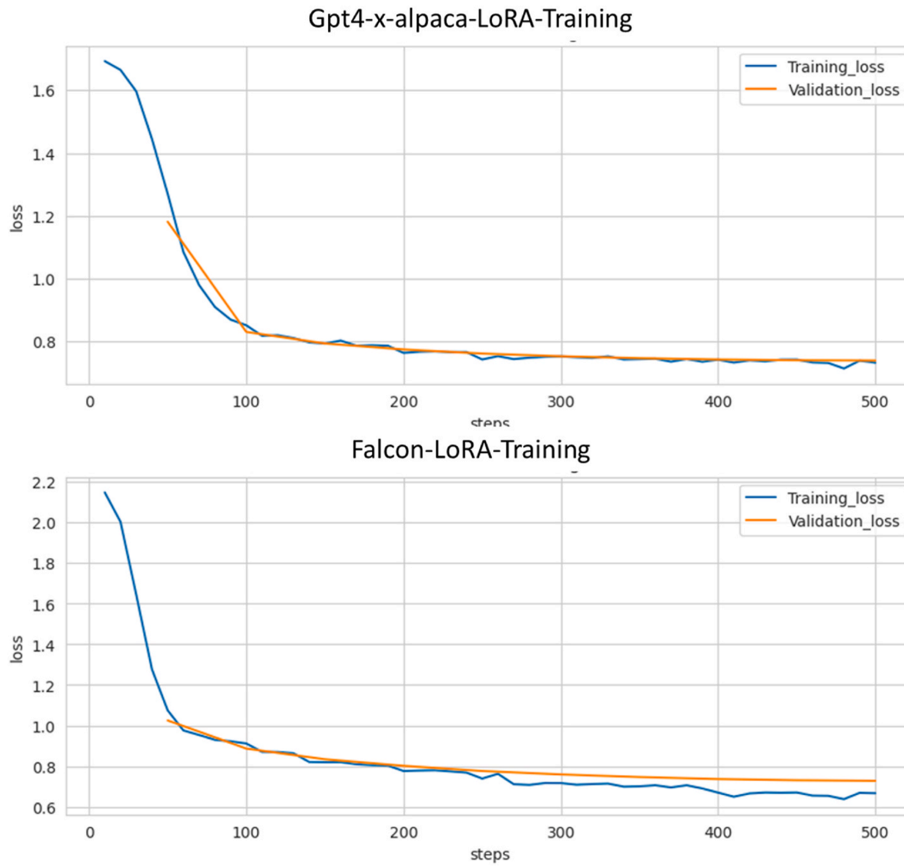

Fig. 2. Training performance of the LoRA training in Module T was assessed. Training for both gpt4-x-alpaca and Falcon models consisted of 500 epochs. For both models, a plateau in validation loss was observed after 500 epochs of LoRA training.

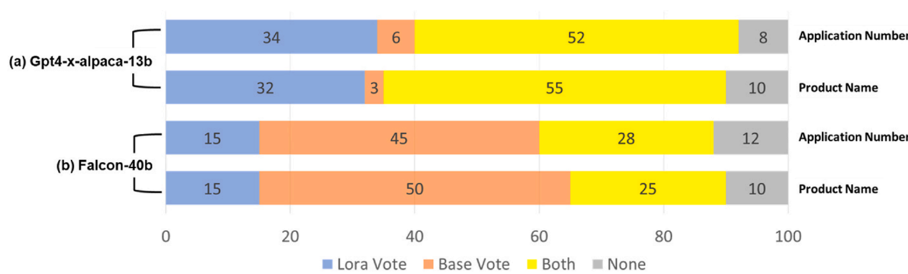

Fig. 3. Comparing LoRA with Base (no-LoRA) in Module T. (a) Comparison for Llama-based models. (b) Comparison between Falcon-based models. All comparisons are judged by GPT-4.

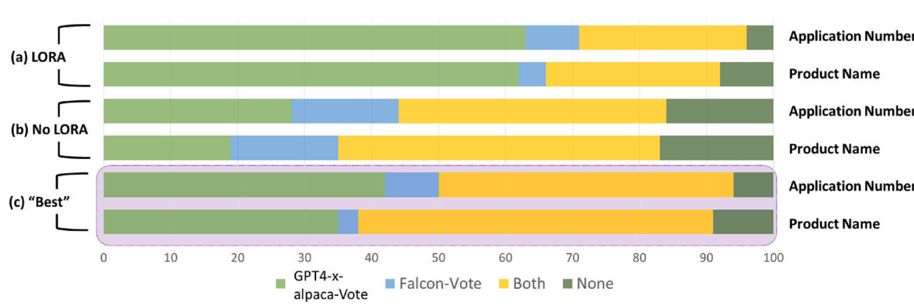

Fig. 4. Performance evaluation of Module T, as assessed by GPT-4, can be summarized as follows: (a) A comparison between LoRA fine-tuned Gpt4-x-alpaca and Falcon model outputs (b) A comparison between raw (i.e., without LoRA fine-tuning); Gpt4-x-alpaca and Falcon model outputs; (c) A comparison between LoRA fine-tuned Gpt4-x-alpaca and raw Falcon model outputs.

models (Gpt4-x-alpaca-13b-LoRA and Falcon-40b) in the same manner, with the results summarized in Fig. 4c. We observed that Gpt4-x-alpaca-13b-LoRA is still a better model, at least as judged by GPT-4. However, over 50% of the answers from Gpt4-x-alpaca-13b-LoRA and Falcon-40b are categorized as both being good, indicating that both models provided good answers to the query.

In all, we demonstrated that for small-size LLMs such as Gpt4-x-alpaca-13b, further fine-tuning with ChatGPT instructed data can improve the model inference performance. On the other hand, for medium or large size LLMs such as Falcon-40b, such fine-tuning process has marginal effect on AskFDALabel's inference performance. It is worth noting that the fine-tuning process we applied only focused on several specific types of queries, and it may affect the model generalization. Therefore, for a specific application, we prefer to use a fine-tuned model such as Gpt4-x-alpaca-13b-LoRA, but for a generalized application scenario, we would use medium-size LLMs such as Falcon-40b, or large-size LLMs such as ChatGPT if the dataset does not have sensitivity concerns.

3.5. Case demonstration: Classifying drugs by cardiotoxicity

To test the framework, we developed a preliminary version of AskFDALabel using Llama-13b-LoRA and tested on classifying drugs by cardiotoxicity based on description of the adverse events in the FDA labeling documents. Our group developed a manually curated database for drug-induced cardiotoxicity (DICT) classification (Qu et al., 2023). This DICTrank dataset classified >1300 human drugs by cardiotoxicity risk using FDA labeling into 4 groups of the cardiotoxicity concern: Most-, Less-, Ambiguous-, and No-DICT-Concern. Here we only used DICT drugs under the Most- and No-DICT-Concern classes. Most-DICT-Concern drugs include those with market withdrawal due to severe DICT, those with DICT events present in the Boxed Warning section of drug labeling, and those with severe or moderate DICT described in the Warnings and Precautions section. No-DICT-Concern drugs are defined as presenting no DICT information in any Adverse Drug Reaction (ADR)-related sections.

Here we asked AskFDALabel the following question: "Does [Drug_Name] contain any cardio-related adverse events or risks?", where the [Drug_Name] is from the list of DICTrank. AskFDALabel returned answers from the top 3 references, and the drug is considered as DICT concern if at least one answer said "Yes". The averaged performance is summarized in Table 2. We also compared the AskFDALabel inference to the ChatGPT result. As shown in Table 2, AskFDALabel outperformed ChatGPT by overall accuracy as 77.65%–71.45%, where ChatGPT was better at Sensitivity and AskFDALabel was better at Specificity.

4. Discussions

In this study, we proposed a framework to integrate LLMs into the local environment to support regulatory application with transparency and trustworthy. This was accomplished by and demonstrated on the FDA labeling documents in a LLM system called askFDALabel. Our study illustrated that it is feasible to use the localized and transparent LLMs tailored to the regulatory needs with customized functionalities. Importantly, our framework could provide a computationally inexpensive solution with comparable performance to support regulatory application only using a small LLM. The shortcoming of a small LLM could be mitigated with a fine-tunning procedure using the domain-specific documents (i.e., FDA Labeling documents in this study). Using this strategy, askFDALabel performed comparable with ChatGPT.

Table 2
DICT drug inference: AskFDALabel and ChatGPT.

DICTrank (Ground Truth)	Model Response	Statistical MEtrics	ChatGPT	AskFDALabel
Most	Yes	**TP**	268	175
Most	No	**FN**	27	113
No	No	**TN**	165	280
No	Yes	**FP**	146	18
		Accuracy	71.45%	**77.65%**
		Sensitivity	**90.85%**	60.76%
		Specificity	53.05%	**93.96%**

4.1. Implementing LLMs with customized needs

AskFDALabel searches the current approved labeling documents and returns the most relevant information about the query question asked by the user. This system uses two recent state-of-the-art LLMs to support generating answers, whose localized implementation allows the user to analyze text documents in a secured environment. In addition, we applied several optimizations including metadata, pattern-match filtering, LoRA fine-tuning, etc., to enhance the capability of AskFDALabel in analyzing labeling documents. By retaining meta-data and the pattern-match filtering, the relevance of the semantic search result was markedly improved. Due to the concern of data accessibility, we only considered localized embedding models rather than embedding methods that rely on an external server, such as the OpenAI embeddings, which could further improve the semantic search performance (OpenAI, 2023).

The full fine-tuning of a pre-train LLM, which retrains all model parameters, is challenge in computational resource and less feasible to many institutes. Using GPT-3 175B as an example – deploying independent instances of fine-tuned models, each with 175B parameters, is prohibitively expensive. We adopted Low-Rank Adaptation, or LoRA, which freezes the pre-trained model weights and injects trainable rank decomposition matrices into each layer of the Transformer architecture, greatly reducing the number of trainable parameters for downstream tasks. As the result, Gpt4-x-alpaca-13b with LoRA fine-tuning was shown to deliver the best performance compared to other LLMs that we have tested, both in result acceptance and inference speed on some common questions used in labeling analysis. Also, compared to direct-use LLMs such as ChatGPT, AskFDALabel is capable of providing references from the original labeling document, as well as being customized to fit a reviewer's needs, such as searching for an Application Number.

However, the current performance of AskFDALabel may still be far from satisfying reviewers and needs more optimization and implementation, even though we have optimized AskFDALabel in many aspects. For example, combining responses from multiple sub-searches to generate a comprehensive answer could be beneficial to help improve the user satisfaction. However, it may also lead to potential information loss and will make reference tracking more complicated. Also, although the overall accuracy of AskFDALabel was better than ChatGPT, it had a higher specificity but lower sensitivity which implied the tool may cause missing result, which is more difficult for reviewers to recover than wrong result (i.e., False Positives). Such preference may be due to insufficient context was provided during the semantic search module. Further improvement is needed to optimize AskFDALabel in this direction.

4.2. Fine-tuning is not always needed and applicable

We observed that Falcon-40b was not benefited from the fine-tuning process under the Alpaca style, which used generated answers from ChatGPT. The advantage of using the Alpaca style of fine-tuning is the efficiency of generating large training datasets (Taori et al., 2023) since ChatGPT can automatically generate answers effortlessly. However, it may not help improve Falcon-40b since Falcon-40b already showed similar power and performance as ChatGPT (Chu, 2023). We found that directly using Falcon-40b without fine-tuning can still provide acceptable results, which provides a potential of directly using some foundational models of LLMs in regulatory application. With that said, it is reasonable to speculate that the good performance of "off-shelf" LLMs on labeling documents might be due to the fact that the labeling documents are already publicly available and similar context such as wiki drug pages may have been "seen" by these LLMs during training. Therefore, caution should be excised when analyzing closed domain data such as regulatory documents that does not readily available in the public domain.

In this study, the fine-tuning process was based the ChatGPT generated datasets. This practice, while common in the community, is inherently depending on the "knowledge" in ChatGPT that is likely lack of information on sensitive data. Therefore, generating a training dataset for applications with sensitive data need to involve and engage reviewer to generate datasets to support the fine tuning of the model. There are several ways to help accelerating the dataset preparation such as reinforcement learning with human feedback (RLHF), using LLMs to generate several candidates and then letting the experts modify based on the best answer, which has been demonstrated to be more efficient than letting the expert write the answer manually (Touvron et al., 2023). Still, this process is quite labor and time intensive. On the other hand, although using larger sized model may not need further fine-tuning, it will also need more computing resources. Based on our experiment, our current computing environment (cluster server with 8 A100 GPUs) can handle 8 gpt4-x-alpaca-13b model or two falcon-40b model in parallel for model inference use. If applying the even larger LLMs such as Llama-70b, only one inference operation may be conducted at a time. It is thus critical to efficiently organize and distribute computing resources within the limited resources. Note that, all model loaded for inference use are not loaded in quantitation mode, by utilizing the quantitation approach such as 8-bit or 4-bit loading, we could potentially increase the number of simultaneous inference calls. This area warrants further exploration for more detailed understanding particularly whether quantitation approach will lead to reduced performance.

4.3. askFDALabel performance on cardiotoxicity assessment

While askFDALabel outperformed ChatGPT on classifying drugs by cardiotoxicity in accuracy, its advantage mainly lies in specificity. We examined these "False Positives" (FPs) and "False Negatives" (FNs) by AskFDALabel. One typical source of FPs is "hallucination". For example, "AMIKACIN SULFATE" has been labeled as a No-DICT concern drug, however, the inference result labeled it as "yes" due to "its potential to induce auditory, vestibular, and renal toxicity, as well as neuromuscular blockade" (FDALabel, 2023). This appears to be an instance of hallucination where the LLM falsely correlated these adverse events to cardiotoxicity, due to the training data it has been trained with. Another potential source of FPs is conflicted answers; for example, one FP answer related to "NITISINONE" is as follows:

> "*Yes, the reference contains information about cardio-related adverse events or risks. The reference mentions that inadequate restriction of tyrosine and phenylalanine intake can lead to elevations in plasma tyrosine levels, which can result in symptoms such as intellectual disability and developmental delay or painful hyperkeratotic plaques on the soles and palms. However, it does not specifically mention any cardio-related adverse events or risks*."

Here, the first part of the answer says "yes" but the second part of the answer says "no," which has confused the AskFDALabel system and resulted in an unclear answer. However, we still observed that AskFDALabel showed a much higher specificity than ChatGPT, which indicates that we can reduce majority of its hallucinations, which is a common issue of current LLMs such as ChatGPT.

With regarding to FNs, one typical source was inaccurate references. For example, "ABACAVIR" is a Most-DICT-concern drug, however, the references provided by the current AskFDALabel engine are texts from labeling sections irrelevant to ADRs, and therefore, the answer returns as "no." According to the low sensitivity we observed from AskFDALabel, there is still a lot of room for improvement in order to get more relevant references in the Module S. Another important factor contributing to the occurrence of FNs in our analysis stemmed from AskFDALabel's inaccurate interpretation of the reference. In several instances, AskFDALabel exhibited the capability to correctly identify and acknowledge pertinent information; however, it failed to understand the information and detect its association with cardiotoxicity. Consequently, AskFDALabel rendered an incorrect "no" response, resulting in FNs. For example, "TRIPTORELIN PAMOATE," a Most-DICT-Concern drug according to DICTrank, was considered to present no cardiotoxicity risk by AskFDALabel based on the following response: "*No. The reference does not specifically mention any cardio-related adverse events or risks. It mentions hypersensitivity reactions, tumor flare, effect on QT/QTc interval, hyperglycemia and diabetes, and cardiovascular diseases, but it does not provide specific information about adverse events or risks related to the cardiovascular system.*" In this case, AskFDALabel adeptly identified the relevant reference and furnished an accurate justification. This answer acknowledged the drug's effect on the QT/QTc interval, which is a common type of cardiotoxicity. However, the model was unable to recognize the association between the effect on QT/QTc interval and DICT, leading to an incorrect interpretation that the drug presented no DICT risk. This suggested that AskFDALabel's comprehension of intricate medical languages, particularly concerning potential adverse effects, may be limited. This limitation underscores the potential for inaccuracies in the model's responses, and it emphasizes the complex nature of drug safety assessments, where a nuanced understanding is essential to prevent FNs.

Additionally, there are instances where AskFDALabel failed to capture crucial information related to cardiotoxicity within the reference. Consequently, it provided an unrelated rationale, resulting in incorrect outcomes. These instances underscore the challenges associated with response accuracy and the potential for misclassification, ultimately contributing to the prevalence of FNs. The full result was provided in Supplementary Table 1.

4.4. Integrating multiple responses into one: an exploration

Based on the current Module S and T, the most relevant text chunks from Module S are passed to Module T separately, and each text chunk would then be used to generate one answer. For example, AskFDALabel generated 10 answers if the top 10 text chunks were processed (Fig. 5a). It is worth noting that Module T may respond like "The adverse reactions of Nymalize are not provided in the given information" if the provided text chunk is not considered to be a valid reference to the query.

We further investigated whether an additional, summarization LLM could be applied to generate a more comprehensive answer (Fig. 5b). As shown, by summarizing the 10 answers from Fig. 5a, a cleaner and more adequate response is returned (i.e., the summarized response). It seems like the summarization model has the capability to filter out invalid answers, as well as rephrase the valid answers into one response. However, we also noticed that the information from the third answer was mostly overlooked, which may cause an information loss during this process. Further investigation will be needed for such potential effect regarding the information loss. It is also important to note that summarizing all answers together may make it more difficult to trace back the original reference of the response. Therefore, in the current version of AskFDALabel, we did not use the summarization model to provide responses.

4.5. A framework rather than a new model

In highlighting the distinctive qualities of our tool, we underscore its nature as a comprehensive and integrated platform. Recent studies have reported that ChatGPT still has improving spaces over answering medical questions (ASHP, 2023), and relatively poor at providing valid scientific references(Gravel et al., 2023; Sanchez-Ramos et al., 2023), both issues further emphasize the importance of providing valid references or

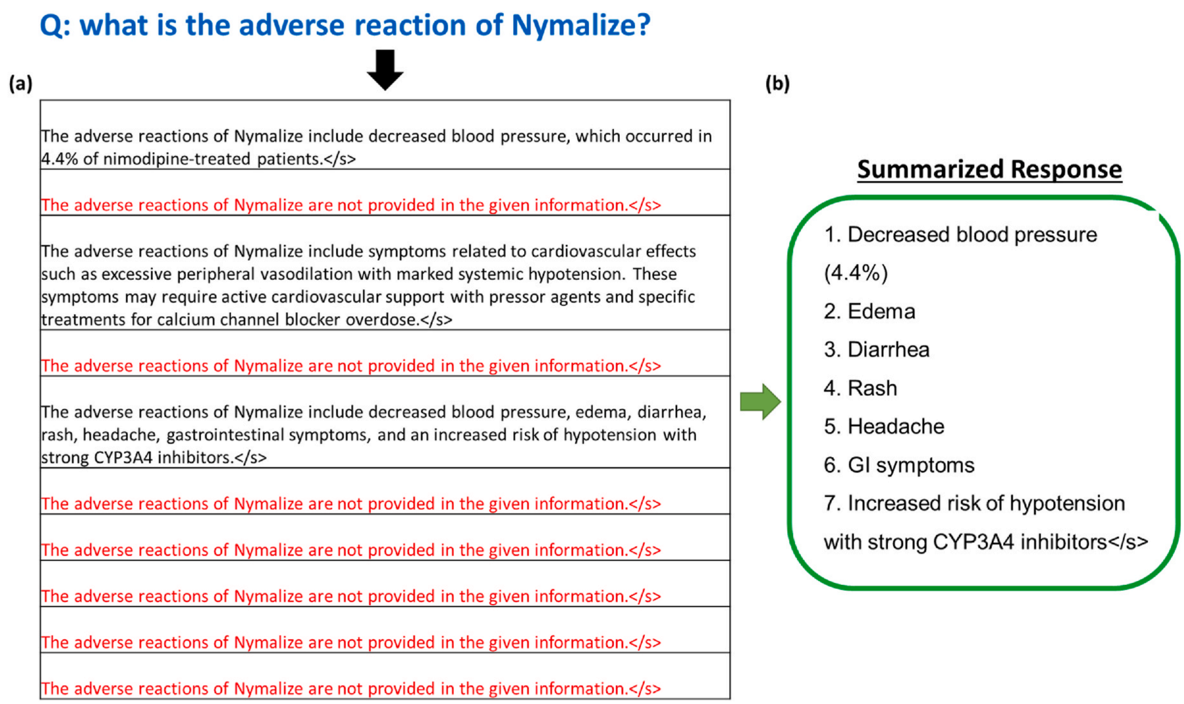

Fig. 5. Summarizing multiple answers into one response using an additional summary LLM.

independent tracing abilities besides the answer generated by LLM. Our platform synergistically amalgamates various functionalities and algorithms, curating an environment that offers a holistic approach to addressing the needs of our users in the realm of regulatory science. Furthermore, it's crucial to articulate that the potential advancements and enhanced local availability of ChatGPT or other LLMs would not render our tool obsolete. On the contrary, any enhancements in ChatGPT or other LLMs, for instance, the newly released GPTs, Llama, Claude, etc., can be leveraged to augment our integrated platform, enhancing its capabilities and performance by harnessing the improved functionality. This symbiotic relationship ensures that advancements in the algorithm not only coexist with our tool but also act as a catalyst for its further development, thereby ensuring its ongoing relevance and utility in an evolving technological landscape.

5. Conclusions

In conclusion, while AskFDALabel presents substantial advancements in the analysis of drug labeling documents, it simultaneously highlights key areas for further research and development. The importance of respecting the confidentiality of sensitive data, optimizing system performance, and accommodating the unique needs of reviewers underscores the complexities of integrating AI within the pharmaceutical domain. Future work should continue to focus on these areas, with the ultimate goal of optimizing the AskFDALabel system to best support the vital work of FDA scientists and reviewers.

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Disclaimer

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CRediT authorship contribution statement

Leihong Wu: Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Joshua Xu:** Writing – review & editing, Supervision. **Shraddha Thakkar:** Writing – review & editing, Project administration, Funding acquisition. **Magnus Gray:** Writing – review & editing, Formal analysis. **Yanyan Qu:** Writing – original draft, Formal analysis, Data curation. **Dongying Li:** Writing – original draft, Formal analysis, Data curation. **Weida Tong:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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